NENA: PROJECT CONCEPT NOTE

Bidirectional Mobile Telephony and Live-Calling Relay Platform for the Deaf Community

PROJECT PROPOSAL DOCUMENT • INNOVATION & SOCIAL IMPACT FRAMEWORK • 2026

1. EXECUTIVE SUMMARY

The **NENA** project is an innovative mobile software platform engineered to dismantle the profound communication barriers separating the Deaf and hard-of-hearing communities from the hearing world across East Africa. Unlike traditional assistive technologies that function strictly as one-way visual guides, NENA introduces a robust, **two-way communication ecosystem** operational across both face-to-face interactions and active telephonic environments (Live VoIP Calling). By combining cutting-edge artificial intelligence—specifically a Few-Shot CLIP Framework for gesture tracking and Whisper AI for automated speech recognition—NENA empowers Deaf individuals to independently navigate hospitals, banks, and markets without the burden or cost of a human sign language interpreter.

2. PROBLEM STATEMENT

Deaf individuals face structural exclusion from basic civic, financial, and medical services due to an absolute breakdown in daily communicative interactions. The primary issues addressed by this initiative include:

- **Inaccessibility of Professional Interpreters:** Relying on third-party human interpreters for daily activities is economically prohibitive and socially limiting for the vast majority of Deaf individuals.
- **Complete Telephonic Exclusion:** Standard telecommunication networks (GSM) rely exclusively on real-time acoustic channels. When a Deaf individual receives a call from a family member, medical practitioner, or financial institution, they cannot accept or respond independently, severely compromising their privacy and autonomy.
- **Low-Resource Data Scarcity:** Tanzanian Sign Language (TSL) and Kenyan Sign Language (KSL) suffer from severely limited annotated digital video corpora. Standard deep learning architectures break down under these data-scarce conditions, demanding millions of training items that simply do not exist.

3. PROJECT OBJECTIVES

1. Develop a cross-platform mobile application (Flutter) that converts Swahili Sign Language into spoken Swahili audio, and incoming Swahili speech into fluent sign language animations in real-time.
2. Incorporate an explicit **User Voice Profile Setting Topology (Option A)** that maps translated text outputs directly to a user-defined gendered voice tracker (Male or Female neural voice profiles) to preserve the user's personal identity and dignity.
3. Implement **Nena Call (In-App VoIP Relay System)**, a live calling interface that converts a hearing caller's voice into sign language visual tokens on the Deaf user's screen, while simultaneously processing the Deaf user's hand gestures through their front camera and streaming neural Swahili speech back to the caller.

Linguistic Resolution: Overcoming Swahili Morphological Constraints

In Swahili grammar, third-person singular pronouns and verbal prefixes are completely gender-neutral (e.g., the sentence "Yeye anakula" maps identically to both "He is eating" and "She is eating"). Because textual tokens parsed from sign language lack inherent semantic gender indicators, Text-to-Speech (TTS) pipelines cannot automatically determine the proper audio pitch profile. NENA gracefully resolves this by decoupling biological detection and instead using an app configuration state variable where users explicitly choose their preferred outbound vocal identity.

4. CORE TECHNICAL ARCHITECTURE & PIPELINES

4.1. Outbound Pipeline: Sign-to-Speech (Expression)

When the user performs sign language in front of the mobile camera, the system passes the visual matrix through a **Few-Shot CLIP-based Contrastive Model**. Instead of matching signs against a massive dataset, the network uses visual-textual alignments to contrast real-time features against a localized dictionary of reference keyframes. The resulting Swahili text string is instantly routed to a localized Neural TTS engine configured to the user's specific male or female profile preference, streaming natural speech to the listener.

4.2. Inbound Pipeline: Speech-to-Sign (Comprehension)

When a hearing individual speaks into the application, an integrated **OpenAI Whisper ASR** model captures the voice data, filtering background ambient noise to output accurate Swahili text. A semantic tokenizer strips away unnecessary filler words to assemble structural keywords. These keywords immediately activate a highly responsive **3D AI Avatar or Video Dictionary** engine on the screen, showing fluent sign expressions that the Deaf user can easily read.

4.3. Telecom Integration: Live Calling VoIP Engine

By leveraging **WebRTC and SIP Signaling Bridges**, NENA creates an active communication loop during telephone calls. To conserve data and battery life, the Deaf user's video feed is processed entirely on the local device's edge hardware. Only lightweight text tokens are passed across the cloud server, where they generate spoken waveforms for the hearing recipient, making the system incredibly stable even over weak 3G networks.

5. IMPLEMENTATION FRAMEWORK

Project Phase	Core Activities & Milestones	Anticipated Deliverables
Phase 1: Data & AI	Collect few-shot calibration video samples for common Swahili signs. Deploy CLIP vision alignment and Whisper ASR blocks inside an accelerated Python workspace.	Functional, low-latency cross-modal translation backend engine.
Phase 2: Frontend	Build the user interface (UI/UX) layout inside Flutter. Establish state management providers to handle the local Male/Female voice profile variables.	Polished interactive mobile app prototype with settings controls.
Phase 3: Telephony	Integrate WebRTC media pathways and hook into Twilio Programmable Voice or SIP gateways to handle real-time remote call data splitting.	Functional in-app live VoIP calling system.
Phase 4: Validation	Conduct active field-testing and focus group evaluations directly within the regional Deaf community to refine accuracy and ergonomics.	Empirically validated, production-ready assistive application.

6. SOCIAL IMPACT & STRATEGIC VALUE

- **Autonomy and Financial Inclusion:** By unlocking independent phone calls, Deaf users gain immediate privacy and safety over medical, personal, and financial communications without third-party exposure.
- **Network Efficiency for Emerging Markets:** Edge processing minimizes high-bandwidth mobile video transmission, allowing users to converse over constrained network data architectures at a minimal cost.
- **Dignified Presentation of Self:** Overriding linguistic limitations via manual app configurations ensures that the translated synthesized speech matches the user's gender presentation and vocal identity perfectly.

7. CONCLUSION

NENA is far more than an isolated utility; it represents a profound, AI-driven bridge for human connection. By uniting sign-to-speech expression, speech-to-sign comprehension, live phone network integration, and customized identity profiles, NENA establishes complete communicative autonomy for millions of underserved Deaf individuals throughout East Africa and globally—permanently turning their mobile devices into reliable pocket interpreters.